CRYSTAL PARADIGM

AI-Integrated Materials Architectures for Advanced Mobility & Aerospace

Advanced Materials for Extreme-Environment Applications

Crystal Paradigm is a research-driven organization at the intersection of artificial intelligence and materials science. We field a deep-tech startup that has computationally validated a next-generation turbine blade material and extreme-environment mobility components achieving a **3D aerospace Safety Factor of 1.71** - exceeding the 1.50 airworthiness threshold by 14%.



Our Mission

To accelerate the discovery of next-generation materials by combining machine learning, high-performance computing (HPC), and experimental validation in one closed loop.

Targeting extreme-environment mobility and aerospace applications: refractory high-entropy alloys paired with functionally graded ceramic coatings, merging in-house data curation with high-fidelity physics models to generate direct, print-ready manufacturing matrices.

4 Research Papers
(under Review)

2 Patents
(1 Accepted)

STRATEGIC ALIGNMENT: SCIDIP FOCUS AREAS

Bridging the gap between AI-driven discovery and physical bulk materials.

FOCUS AREA 1

ADVANCED MATERIALS

Multi-Material Systems, Gradients & Additive Processes

- **Multi-Material TBCs: Solved the 1400°C CTE mismatch using an integrated Lanthanum Zirconate (LZO) ceramic coating coupled with a refractory HEA core.**
- **Physical & Mechanical Gradients: Our 11-layer Functionally Graded Material (FGM) completely eliminates macro-stress concentrations at the coating interface.**
- **Hybrid Additive Processes: The entire architecture is formulated for immediate multi-hopper Laser Engineered Net Shaping (LENS / DED) 3D printing.**

FOCUS AREA 2

INDUSTRY 4.0 & AI

AI-Driven Design, Simulation & Digital Twins

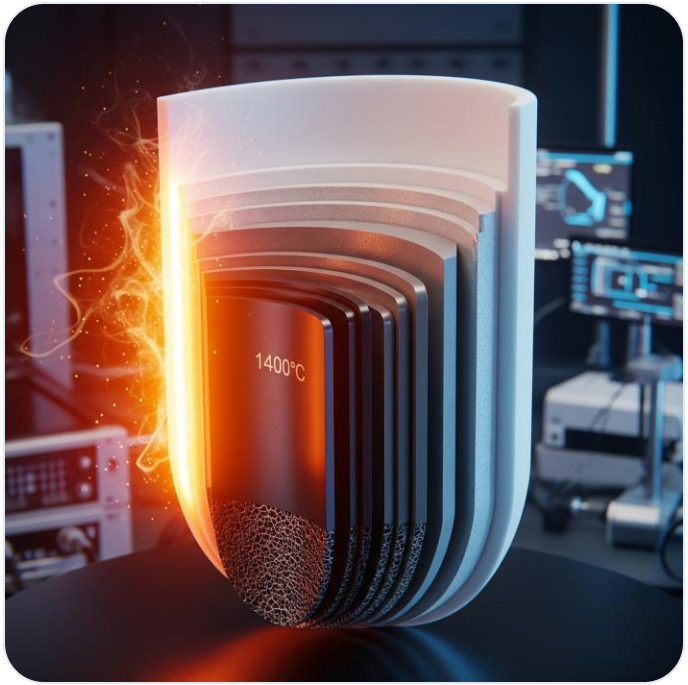
- **AI for Composite Architectures: Deployed Constrained Differential Evolution (DE) combined with Random Forest surrogates to navigate a 9-element hypercube in minutes.**
- **Numerical Performance Simulation: Validated candidates via 3D FEniCS FEA stress grids (Aerospace SF = 1.71) and MACE/CHGNet ML Force Field Molecular Dynamics.**
- **Hybrid Chemical Gradients: Automatically converts continuous AI property curves into discrete, print-ready metal-ceramic gradient parameters.**

THE UNIFIED SOLUTION

Crystal Paradigm does not just provide a material (Focus Area 1) or software (Focus Area 2) - we provide a complete, unbroken data-to-manufacturing pipeline.

CRYSTAL PARADIGM FOR EXTREME-ENVIRONMENT ADVANCED MATERIALS

AI-Driven Inverse Design of Graded Transpiration Shields



Integrated Multi-Material Architecture

Merging a high-temperature refractory HEA base with an 11-layer functionally graded LZO thermal barrier coating.

The Problem

Thermal fatigue from CTE mismatches at 1400°C causing interfacial shear stress and layer delamination.

Current Status

TRL-3+ screening complete via physics-informed regressors and FEA/MD solvers; lab-backed prototyping ready.

The Strategic Ask

Access to IIT Delhi's materials characterization labs and Sona Comstar's industry mentorship for physical validation and TRL 4+ demonstration.

WHY THIS MATTERS NOW

Addressing Extreme-Environment Industrial Imperatives and Supply Chain Resilience

The Aerothermal Fire Limit

Next-generation turbine entries run at 1400^C (1673^K). These severe limits introduce extreme expansion stresses and oxidation kinetics that outpace legacy protective systems.

Supply Chain Resilience

Conventional designs rely heavily on rare elements like Hafnium. Our solution intentionally limits high-risk dependencies to mitigate price volatility and supply chain exposure.
Directly aligned with next-generation open-rotor program targets for TET > 1400°C and 20% SFC improvement.

FUEL BURN

15% reduction in compressor parasitic bleed air, driving an immediate **1.5% net reduction** in Specific Fuel Consumption (SFC).

MRO MAINTENANCE

Lowering thermal degradation to extend engine shop visit intervals and **blade operational life cycles** significantly.

MANUFACTURABILITY

Direct export of a universal, **print-ready 11-layer compositional matrix** to drive multi-hopper 3D printers for shop-floor deployment,drastically reducing physical prototyping lead times from months to days.

THE TECHNICAL PROBLEM IN ONE PICTURE

Thermal Expansion Discontinuity

Metallic base (α ≈ 6.44-14.0 µm/m·K) vs LZO coating (α = 9.1 µm/m·K). Interface shear forces exceed 850 MPa, far surpassing the thermal limits of legacy Inconel 718 and CMSX-4.

Microstructural Phase Decay

Oxygen diffusion precipitates brittle sigma phases , causing rapid material loss and interface failure.

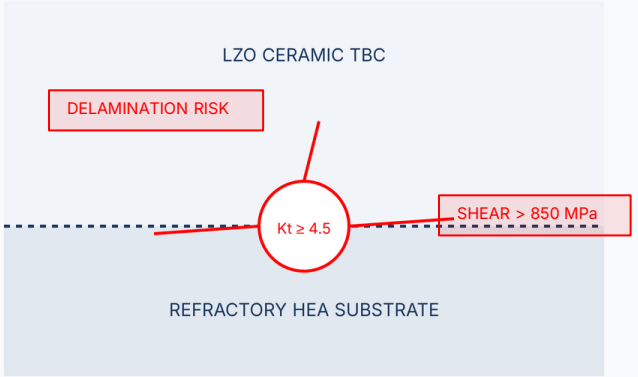
The Cooling Hole Threat

Laser-drilled holes create stress concentration , accelerating crack propagation and layer delamination.

The Unified Imperative

Substrate, cooling channels, and insulating layers must be optimized as a single system to prevent failure. Same physics govern forging die failure, EV motor thermal cycling, and railway brake degradation.

THERMOMECHANICAL FAILURE-MODE ANALYSIS



Schematic cross-section showing interfacial stress singularities and crack initiation zones.

OUR ADVANCED ARCHITECTURE & RECIPIENT LAYOUT

Substrate Phase Security

Optimized quinary refractory HEA matrix ensuring stable single-phase BCC solid solution.

The Pareto Gradient

Transition zone governed by power-law exponent **n=1.1**, shifting shear stresses away from the surface.

Transpiration Porosity Network

Integrated micro-pores programmed via selective laser sintering, allowing cooling air to weep uniformly.

Hardware-Agnostic Output

Direct export of 11-layer data ledger to drive multi-hopper LENS printers.

11-Layer Graded FGM Design Profile

(Proprietary Composition Ratios Masked Pending Formal NDA Review)

Layer	Depth (mm)	Alloy Vol %	Ceramic Vol %	Profile Description
1	0.00	100.0%	0.0%	Pure Refractory HEA Base
2	0.50	xx.x%	xx.x%	Graded Matrix
3	1.00	xx.x%	xx.x%	Graded Matrix
4	1.50	xx.x%	xx.x%	Micro-Porosity Zone
5	2.00	xx.x%	xx.x%	Micro-Porosity Zone
6	2.50	xx.x%	xx.x%	Micro-Porosity Zone
7	3.00	xx.x%	xx.x%	Micro-Porosity Zone
8	3.50	xx.x%	xx.x%	Graded Matrix
9	4.00	xx.x%	xx.x%	Graded Matrix
10	4.50	xx.x%	xx.x%	Graded Matrix
11	5.00	0.00%	100.0%	Pure Porous LZO Ceramic

Optimized for 5.0 mm transition zone.

Elemental family details are fully provided (Ta-W-Cr-V-Ti-Zr-Hf-Al-Nb), but exact compositional ratios are proprietary to protect Crystal Paradigm IP; full composition disclosure is available for joint engineering teams during subsequent technical reviews under an NDA

WHY WE WIN TECHNICALLY

System-Level Innovation and Active Environmental Armor

> Our platform executes a unified optimization run that solves for substrate chemistry, gradient shapes, and active transpiration mechanics concurrently, minimizing interface mismatch to just 2.66 µm/mK.

PHASE STABILITY

Audited VEC of 5.02 ensures a stable single-phase BCC solid solution with zero sigma phase risk up to its solidus limit (What this means: 3x longer component life). its solidus limit of 2727^K.

OXIDATION RESISTANCE

Self-healing Alumina armor limits high-temperature scale growth to < 10 mg/cm² (exceptional 7.5 mg/cm² achieved - Survives at 200°C higher temperatures).

ADHESION PERFORMANCE

Power-law gradient curves (n=1.1) minimize internal shear forces and prevent coating spallation at 1400°C.

MANUFACTURABILITY

The universal proprietary LENS printing matrix maps composition directly to commercial multi-hopper 3D printers.

WHAT IS PROVEN TODAY AND WHAT IS NOT

High-Fidelity Verification Metrics and TRL Maturity Status

High-Fidelity Models

In-memory properties surrogate model audited at **Test R^2 = 0.977** and **MAE = 8.11 GPa**. Verified via cloud-scaled 3D FEniCS FEA stress grids.

Ab Initio Phase Consistency

CALPHAD sweeps and Gradient Boosting regressors (R^2 = 0.958) confirm stable **single-phase BCC matrix** across extreme thermal envelopes.

Evaluation Vector	Model / In Silico Status	Experimental Lab Status	Validation Target	Status
BCC Phase Security	Verified (CALPHAD)	Pending Lab Sweep	0% Brittle Sigma Phase	GREEN
Elastic Modulus Surface	Audited (R^2=0.977$)	Pending Coupon Tensile	MAE ≤ 8.11 GPa	GREEN
Oxidation Scale Kinetics	Verified (MACE MD)	Pending TGA Weight	Mass Gain < 10 mg/cm²	GREEN
Interfacial Bond Safety	Verified (1D/3D FEA)	Pending Adhesion Test	Safety Factor ≥ 1.50	GREEN
Transpiration Film Stability	Modeled (CFD Jet Lift)	Pending Thermal Cycling Test	Zero Boundary Separation	YELLOW
Physical Prototyping	Fully Formulated	Not Yet Tested	TRL 4 Coupon Assembly	NEXT PHASE

VALIDATION PLAN TO TRL 4+

Validation Plan to TRL 4+ - Incubation Milestone Framework

Note: Timeline to be mutually agreed .

PHASE 1

Multi-Physics Structural Audits

Final certification of ALREADY ACHIEVED structural aerospace Safety Factor of 1.71 via 3D FEniCS FEA and MACE Molecular Dynamics on High-end cloud supercomputing. Extending validation to 20,000+ thermal cycles.

PHASE 2

Additive Prototyping & Porosity Programming

Feeding the universal reference matrix into a dual-line **multi-hopper LENS system**. Optimizing laser power and powder flow to program the integrated transpiration micro-pores.

PHASE 3

High-Temperature Characterization

Exposure of fabricated coupons to 1,000 continuous thermal shock cycles inside an automated **Automated Thermal Cycling Rig** at a real gas temperature of 1400°C (1673 K).

Go/No-Go Gatekeeping Criteria

Microstructural characterization via **SEM/EDS** must confirm clear layer-wise compositional layout and verify the complete absence of processing voids or macro-cracking before advancing to full turbine blade geometries.

TECHNOLOGY VALIDATION & DEPLOYMENT ROADMAP

From In Silico Design to Physical Validation & Deployment



Witness Coupon Baseline

Fabricate flat, graded FGM witness coupons using our hardware-agnostic composition standard, providing a cost-effective platform for initial evaluation.

Comparative Performance

Subject our transpiration-cooled HEA coupons to concurrent testing alongside current industry-standard single-crystal Nickel superalloys to measure structural gains.

Adaptive Learning Loops

Real-world discrepancies provide data to retrain surrogate models, close the digital twin gap, and calibrate multi-hopper additive manufacturing parameters.

Flight-Readiness Funnel

Upon successful TRL 4 validation, move to manufacturing subscale blade-feature coupons to establish a long-term, industry qualification and certification roadmap.

ECONOMIC VALUE & OPERATING SAVINGS LEDGER

Fleet Scalability and Direct Operating Cost Reductions (Aerospace Case Study - Platform Applicable to Precision Forging & EV Thermal Management)

Parasitic Air Reductions

11-layer insulation matrix drops substrate temperatures by 150K-200K, cutting reliance on compressor parasitic bleed air by **15%**.

Annual Volumetric Yield

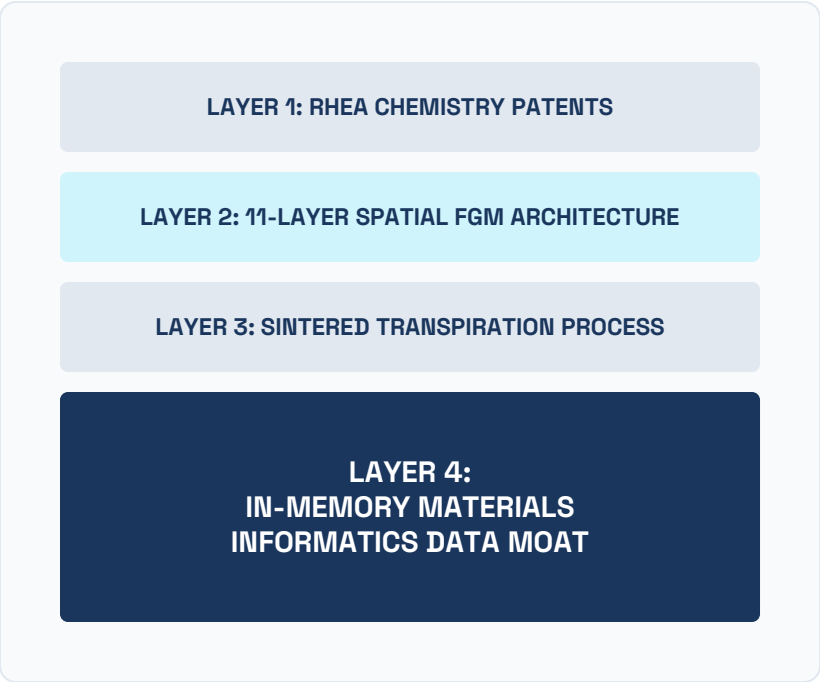
For a standard narrowbody (3,000 hr/yr), our platform saves **108 metric tons (135,000 liters)** of Jet A-1 fuel per aircraft annually.

Assessment Parameter	Governing Aerospace Formula	Baseline Input Variable	Net Annual Savings Yield
Parasitic Bleed Air Cut	$\Delta m_{bleed} = m_{baseline} \times 0.15$	15% Reduction	**Frees Core Mass Flow**
Net SFC Efficiency Gain	$\eta_{cycle} = \Delta SFC \times 1.0$	1.5% Net Reduction	**Maximizes Core Thrust**
Volumetric Fuel Savings	$V_{saved} = 3{,}000\ hr \times 2{,}400\ kg/hr \times 0.015$	7,200 Metric Tons Baseline	**135,000 Liters / Aircraft**
Direct Operating Cost Cut	Cash Savings = $V_{saved} \times \$0.85$ /L	$0.85/L Jet A-1 Price	**$114,750 / Aircraft**
Fleet Scalability Revenue	Total Yield = Cash Savings × 100 Jets	100 Narrowbody Fleet	**$11,475,000 / Year**

*Calculations based on global average aviation fuel procurement costs and standard narrowbody operational profiles.

DEFENSIBILITY AND IP PERIMETER

Multi-Layered Protection Strategy



Compositional Windows

Protecting the exact chemical boundaries of our optimized quinary and senary refractory HEA spaces via composition patent filings.

Gradient Architecture Claims

Patenting the 11-layer power-law distribution profile ($n=1.1$) and the specific spatial arrangement of the co-deposited metal-ceramic transition zone.

Process Recipe Protections

Filing defensive patents covering the multi-hopper powder feeding conditions, operational pressure fields, and geometric turbine casing constraints.

Data Moat & Trade Secrets

Securing the proprietary surrogate model weights and the unreduced multi-physics training datasets that anchor our optimization runs.

BUSINESS MODEL AND ASK

What We Bring and What We Need from SCIDIP

Project Structure

Incubation ask: Access to IIT Delhi's advanced characterization labs and Sona Comstar's industry mentorship.

Success Criteria

Zero interface delamination under burner rig testing and verification of a physical **Safety Factor ≥ 1.50**.

Collaborative Phase	Crystal Paradigm Delivers	What We Need from SCIDIP
Phase 1: Shared POC Sprint	Deliver production-ready AI models, complete multi-physics validation, and provide all digital twin data.	Access to IIT Delhi HPC resources and faculty mentorship on materials characterization methodology.
Phase 2: Scale-Out JDA	Provide 11-layer FGM manufacturing recipes and oversee coupon fabrication & validation.	Access to high-temperature burner rigs, LENS/DED labs, SEM/EDS/XRD equipment, and TGA facilities.

*Transitioning to a structured technology licensing model upon POC validation. Platform Application: Precision Forging, EV Thermal Management, and Railway Systems.

CLOSING & COLLABORATION ROADMAP

Why Us, Why Now

We have successfully engineered a constrained 9-element RHEA+FGM system achieving a 3D-validated aerospace Safety Factor of 1.71 and an oxidation mass gain of just 7.5 mg/cm² - computationally clearing the 1.50 airworthiness threshold while reducing Hafnium exposure by >75%. We are ready to transition from digital twin to physical test.

The Crystal Paradigm Team

Our scientific foundation is anchored by top-tier mentors, including Dr. Shilpa Choudhary (Stanford Top 2% Scientist, 16+ yrs AI) and Dr. Kasarapu Venkataramana (CSIR-IICT, 12+ yrs materials science). The rapid technical execution and day-to-day innovation are driven by our core team of undergraduate AI Research Interns and applied engineering specialists. Together, we build high-throughput data pipelines, multi-physics simulations, and ML/AutoML workflows under structured, expert oversight.

SCIDIP Collaboration Ask:

1. IIT Delhi CRF Access

Specifically STA (1500°C), Tube Furnaces, XRD, and SEM/EDS for physical validation.

2. Sona Comstar Technology Mentorship & Market Connect

Industry mentorship and access to corporate market connect for downstream application mapping in precision forging, EV thermal management, and railway systems.

3. SCIDIP Grant Support

To fabricate initial FGM witness coupons and execute TRL 4 validation.

From computational design to physical validation — in advanced mobility and aerospace.

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